

Community Assembly II

Today's Agenda:

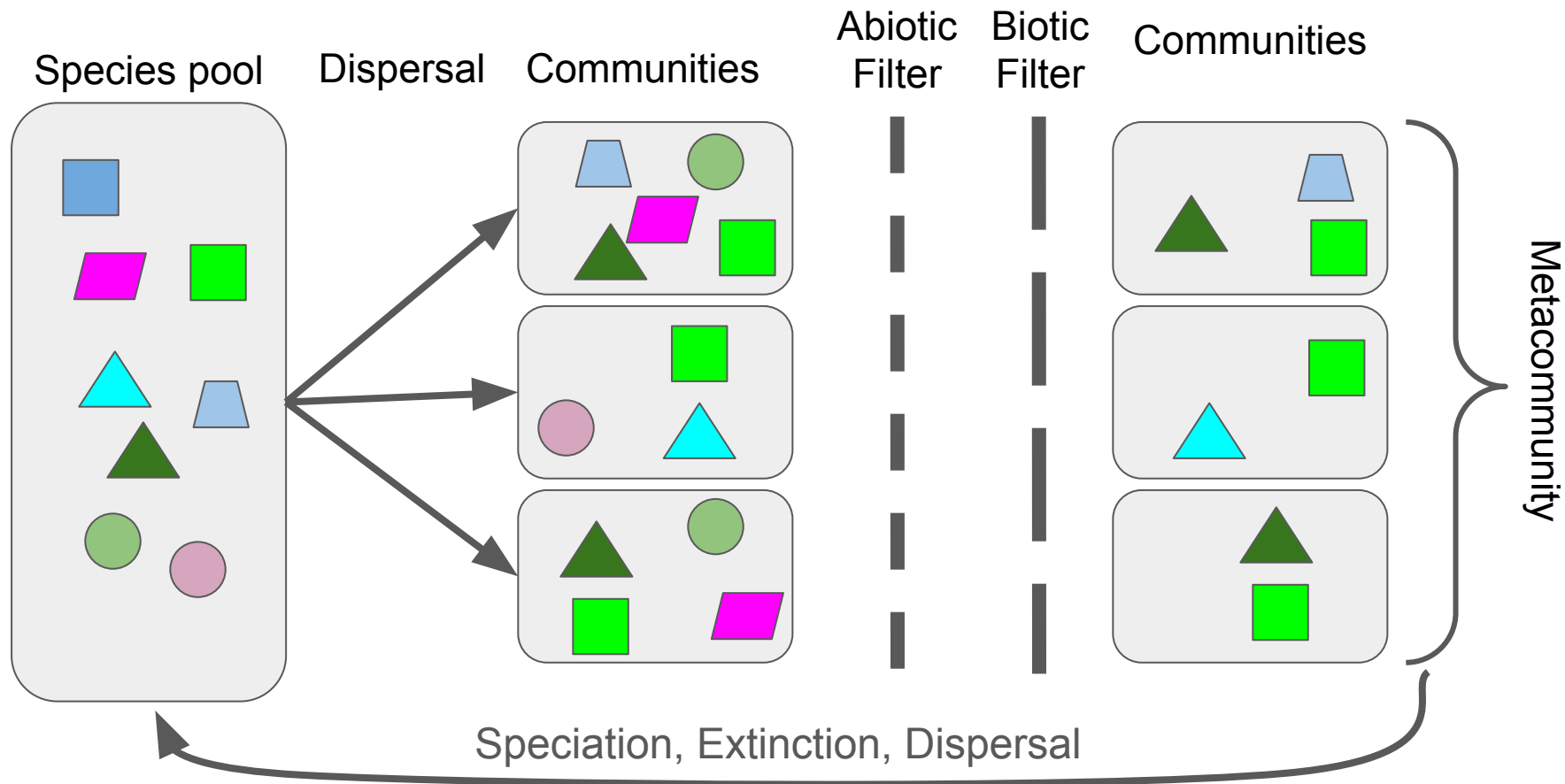
- Quiz
- Check-in
- Community Assembly Examples

Check-in

Just wanted to touch base and see how the class is going.

Anything that isn't working? Anything that IS working?

Feel free to email me as well.



What does this look like in practice?

Environmental Filtering of Plants

Studied a chronosequence of soil ages

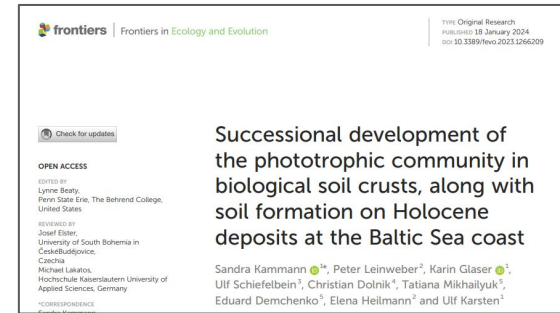
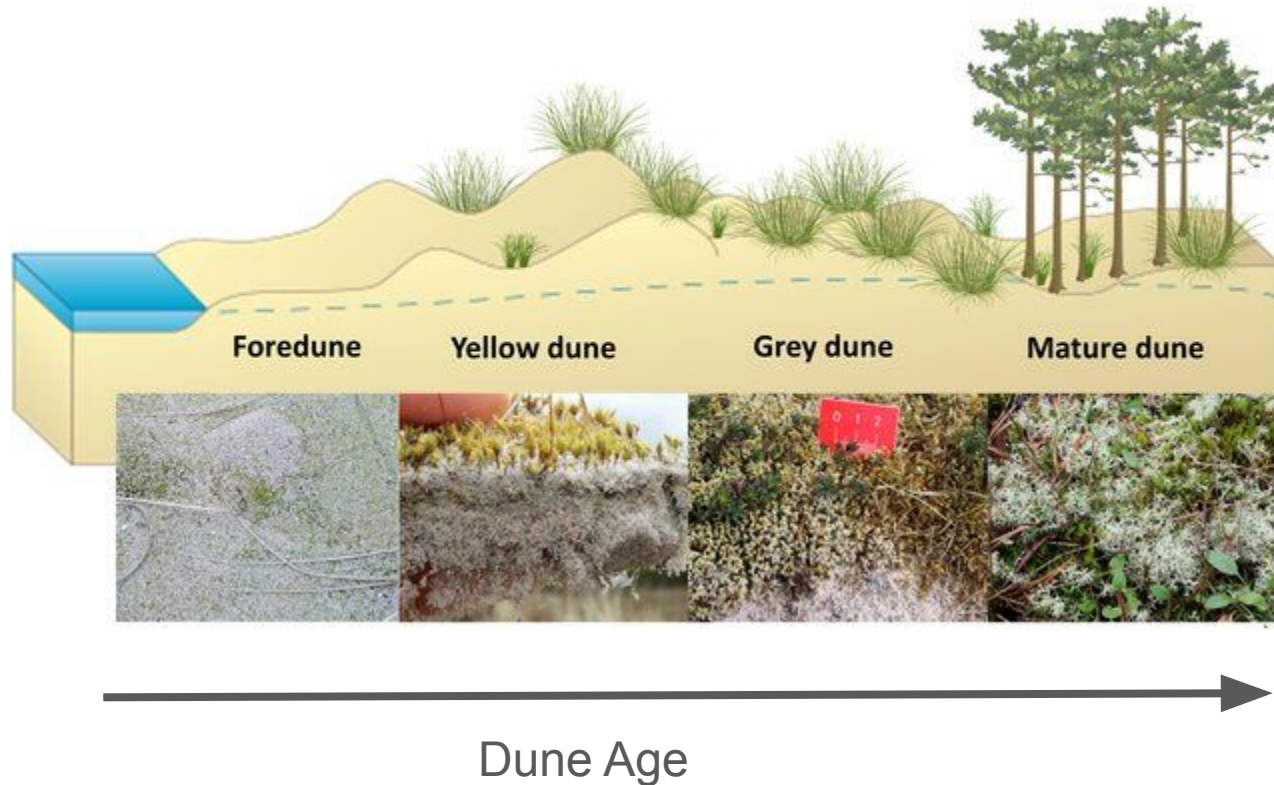
- “Space-for-time” substitution
- Dunes in Australia

PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

Etienne Laliberté,^{1*} Graham Zemunik,¹ Benjamin L. Turner²

Aside: space for time substitutions



- Note: this figure depicts Baltic dunes

Environmental Filtering of Plants

Studied a chronosequence of soil ages

- “Space-for-time” substitution
- Dunes in Australia

Really cool system!

- 2 million year gradient of soil ages
- Sites within 10 km (assume the same regional species pool)
- Comparable climate
- Open ecosystem (not light-limited)

Environmental filtering explains variation in plant diversity along resource gradients

Etienne Laliberté,^{1*} Graham Zemunik,¹ Benjamin L. Turner²

Environmental Filtering of Plants

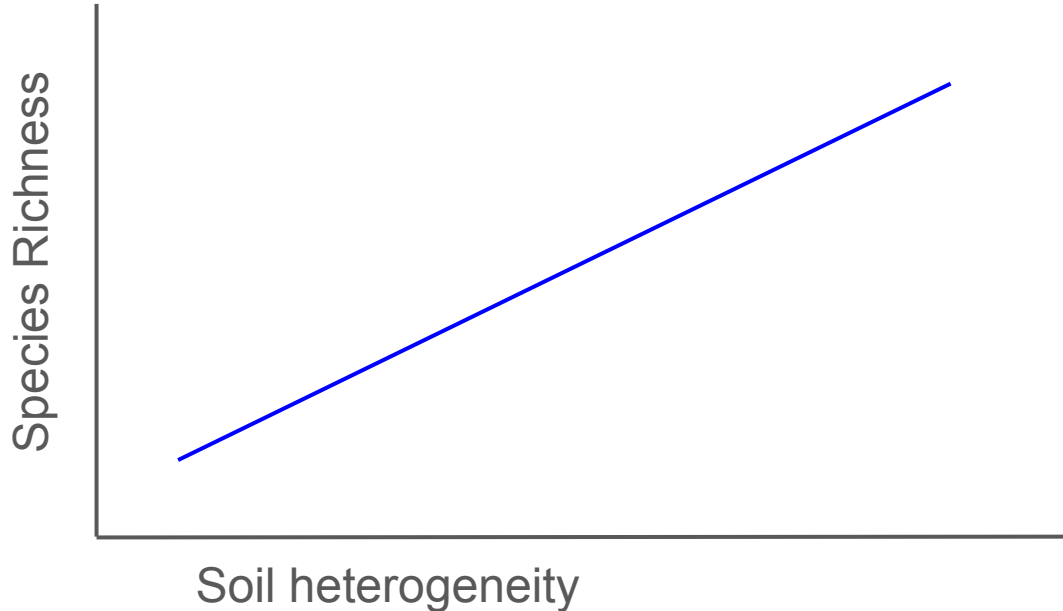
PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

Etienne Laliberté,^{1*} Graham Zemunik,¹ Benjamin L. Turner²

Hypotheses tested

More heterogeneous soils, more species - why?



Environmental Filtering of Plants

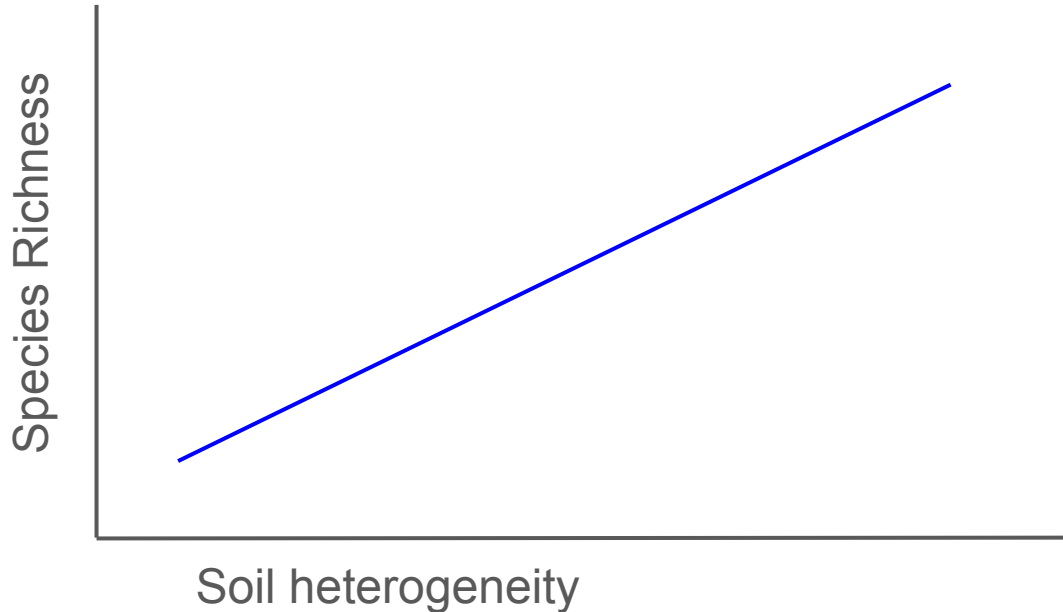
PLANT ECOLOGY

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Hypotheses tested

More heterogeneous soils, more species (niche partitioning)



Environmental Filtering of Plants

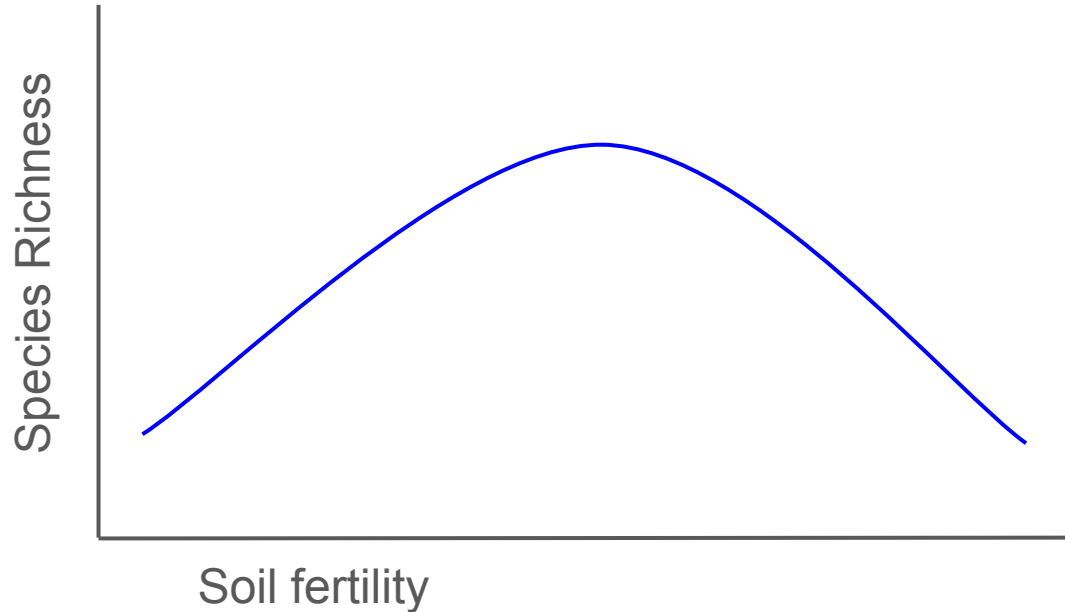
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Hypotheses tested

Unimodal response with soil fertility - why might we expect this?



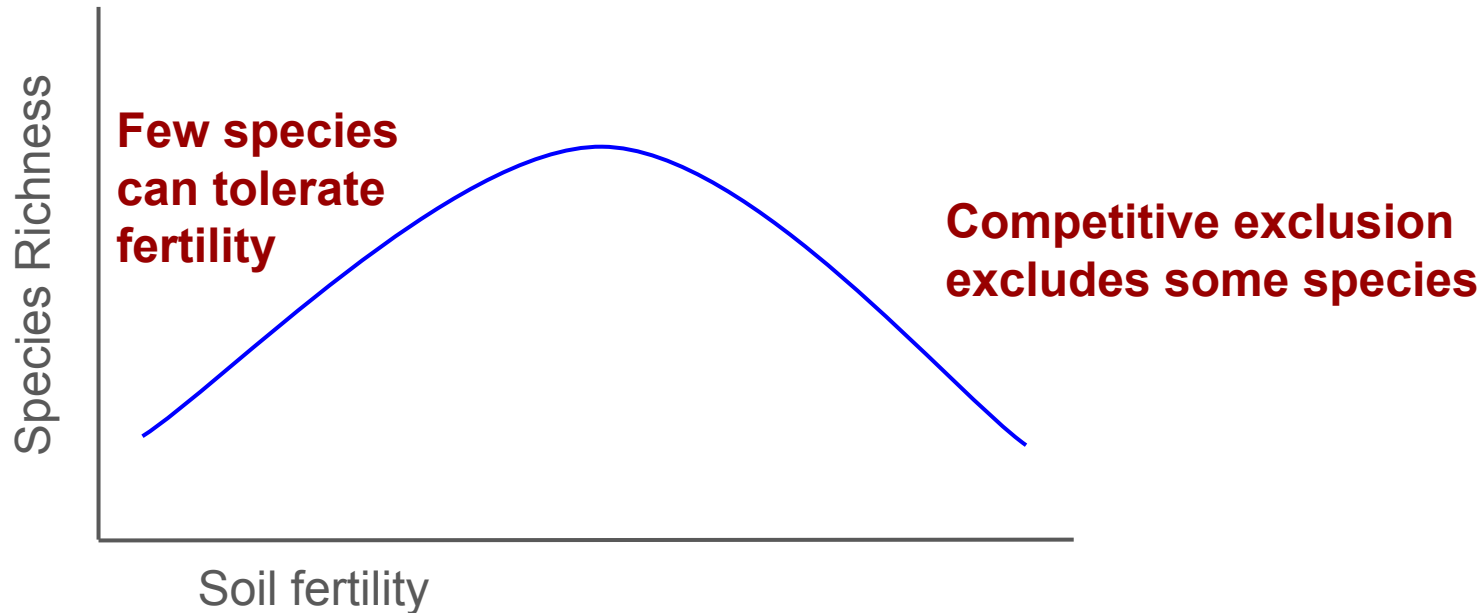
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Environmental Filtering of Plants

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Environmental Filtering of Plants

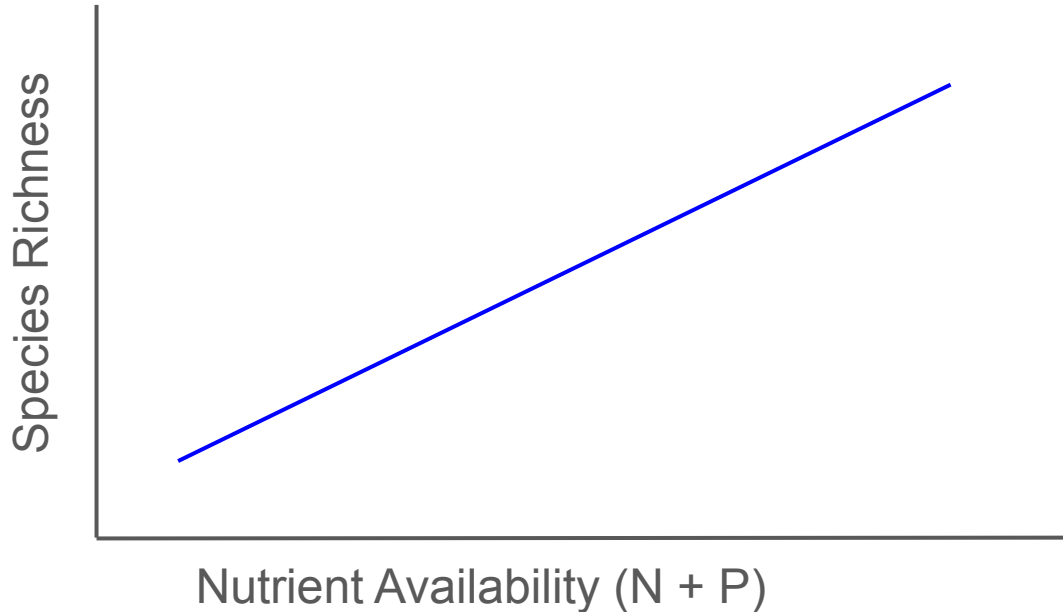
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Hypotheses tested

Higher diversity where nutrients aren't limiting



Environmental Filtering of Plants

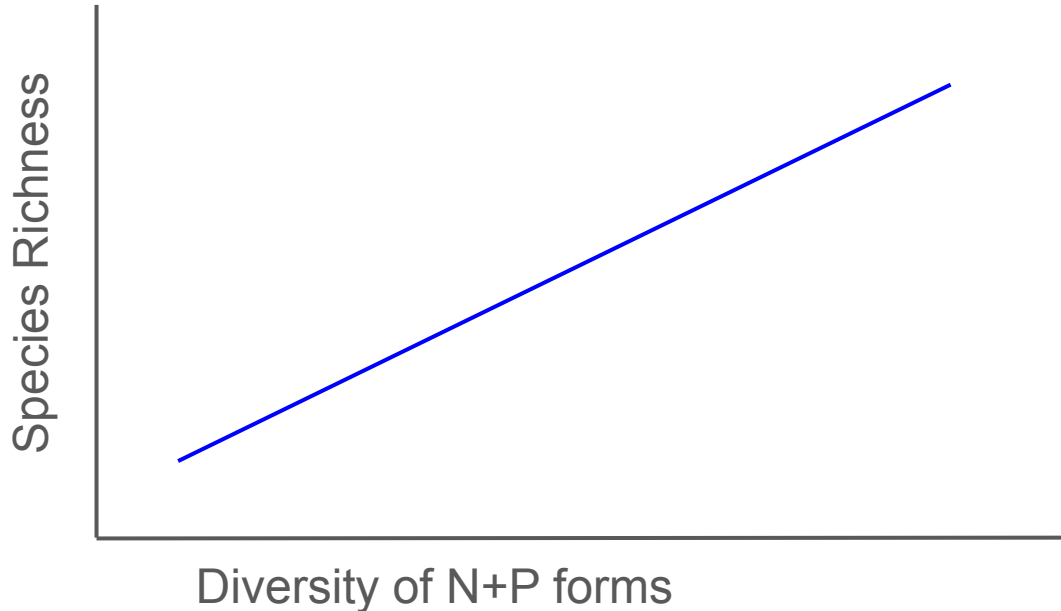
PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

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Hypotheses tested

Higher diversity where multiple forms of nutrients exists (partitioning)

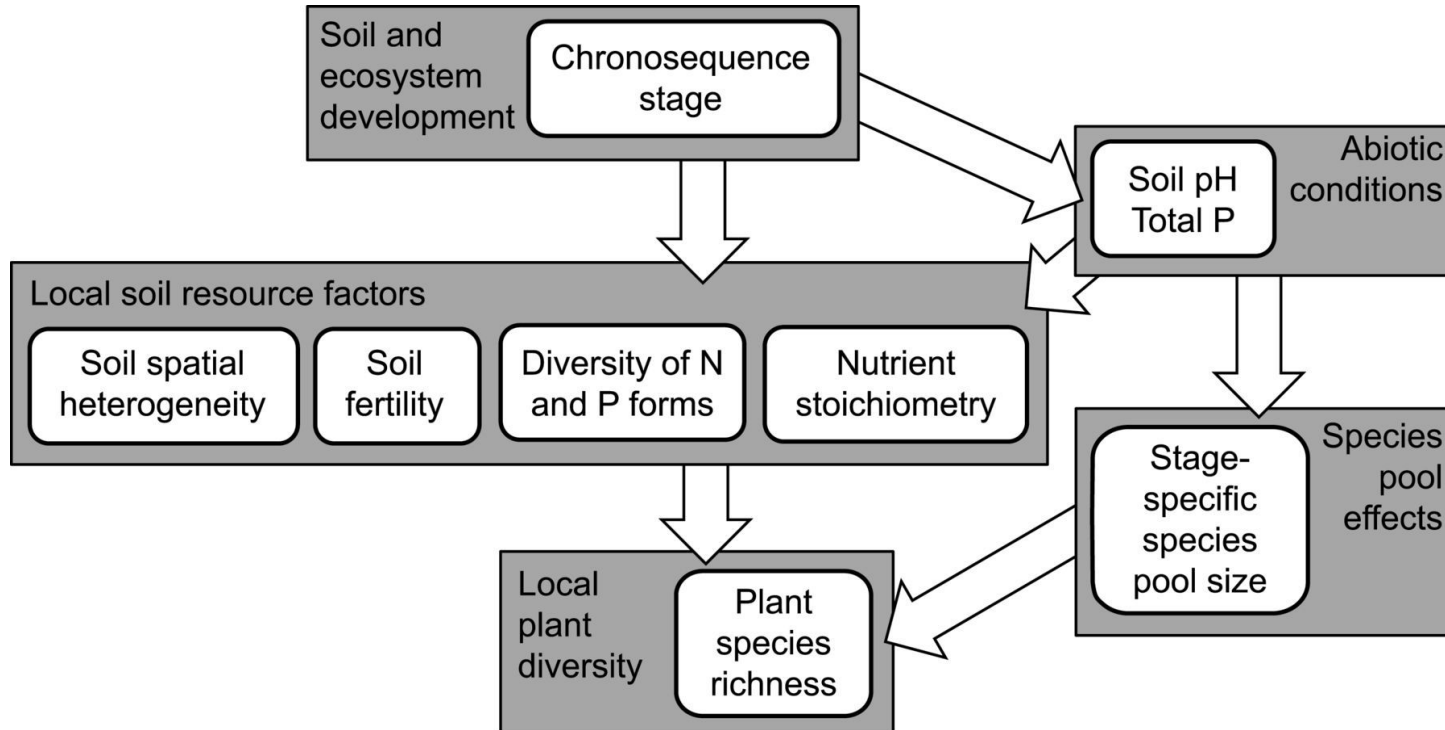


Environmental Filtering of Plants

PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

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Environmental Filtering of Plants

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Methods

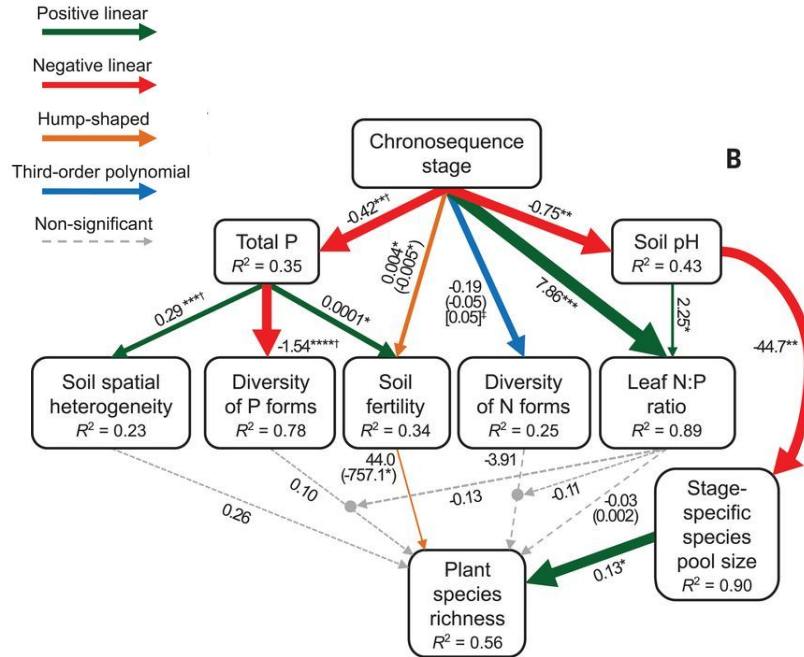
- 60 plots (10 m x 10 m)
- 6 age classes (10 plots each)
- In each plot
 - Species
 - Density
 - Cover
 - Soil samples (nutrients and fertility analyses)
 - Leaf sample (for nutrient analyses)
- Species pool estimated:

$$S_{\text{Jack2}} = S_{\text{obs}} + \left[\frac{Q_1(2m-3)}{m} - \frac{Q_2(m-2)^2}{m(m-1)} \right]$$

where S_{Jack2} = extrapolated richness, S_{obs} = observed richness, Q_1 = number of species found in only one sample, Q_2 = number of species found in precisely two samples, and m = number of samples. Nonparametric richness estimation was performed using the 'vegan' packages in R (48).

Environmental Filtering of Plants

Results



Environmental filtering explains variation in plant diversity along resource gradients

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Environmental Filtering of Plants

PLANT ECOLOGY

Environmental filtering explains variation in plant diversity along resource gradients

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Plot-level properties didn't explain plot richness very well

Instead:

- Soil aging changed soil pH

- Different pools of species were specialized on different pH values

Competition and ciliates

LETTER

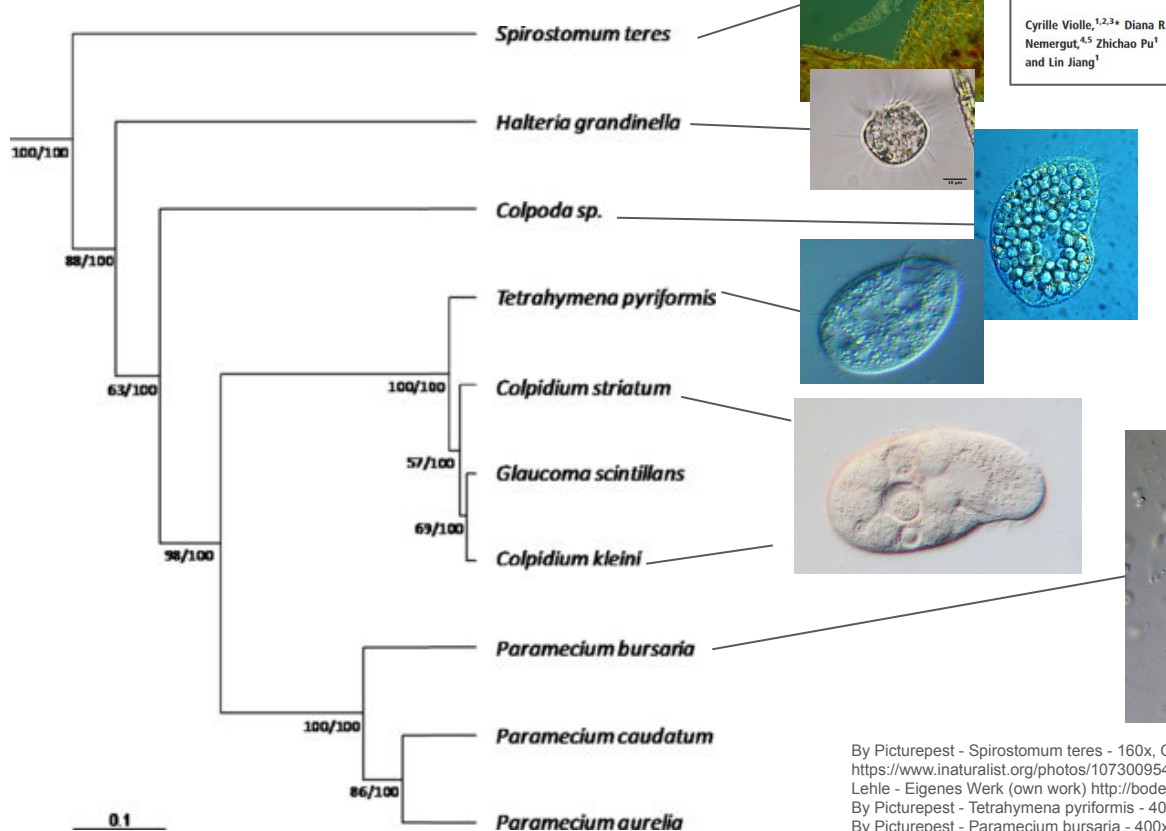
Phylogenetic limiting similarity and competitive exclusion

Cyrille Violle,^{1,2,3*} Diana R.
Nemergut,^{4,5} Zhichao Pu¹
and Lin Jiang¹

Abstract

One of the oldest ecological hypotheses, proposed by Darwin, suggests that the struggle for existence is stronger between more closely related species. Despite its long history, the validity of this phylogenetic limiting similarity hypothesis has rarely been examined. Here we provided a formal experimental test of the hypothesis using pairs of bacterivorous protist species in a multigenerational experiment. Consistent with the hypothesis,

Competition and ciliates



LETTER

Phylogenetic limiting similarity and competitive exclusion

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Cyrille Violle,^{1,2,3*} Diana R. Nemerut,^{4,5} Zhichao Pu¹ and Lin Jiang¹

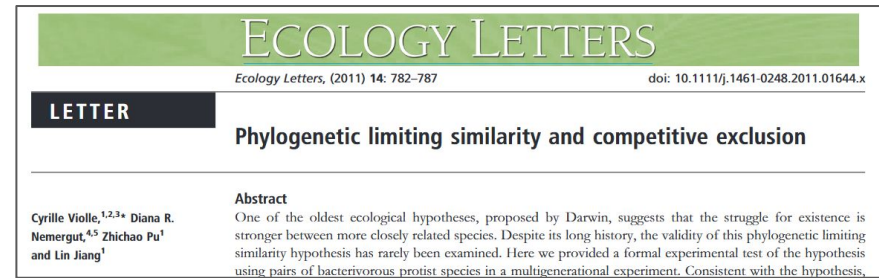
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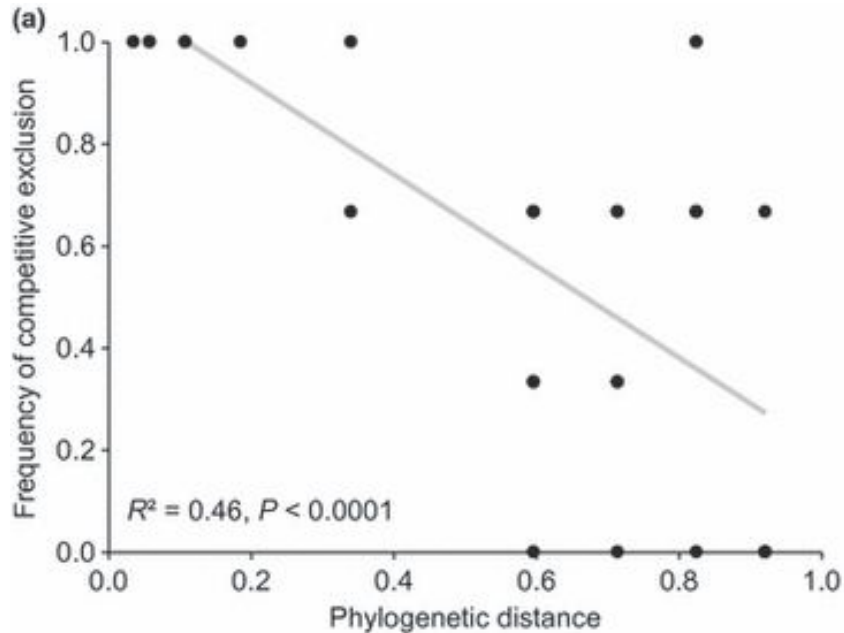
Competition and ciliates

Methods

- Grew populations in pairs and alone
- Grown with 3 different bacteria species (= food)
- Tracked population sizes over time



Competition and ciliates



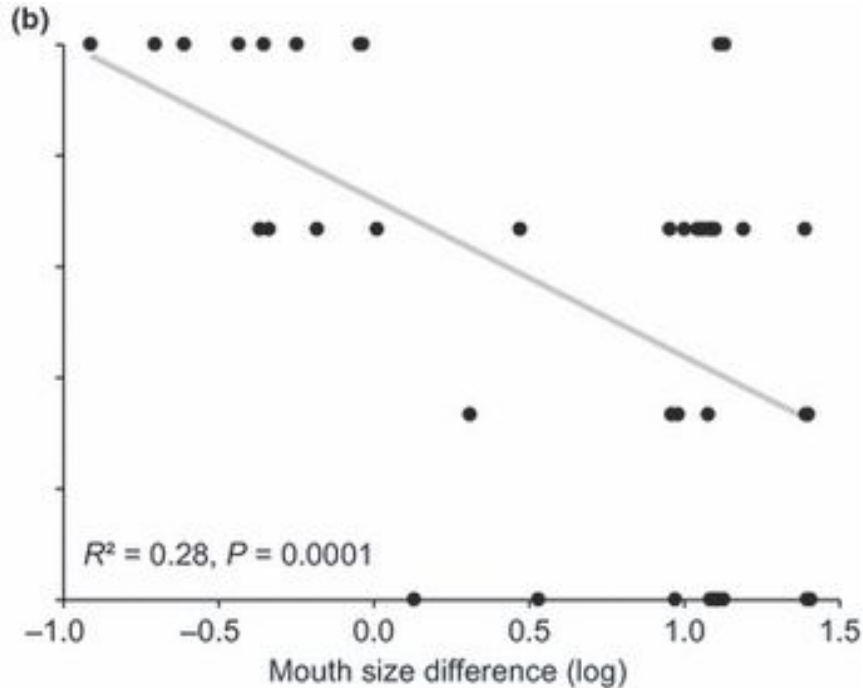
Close relatives less likely to coexist -
suggests competitive exclusion

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Competition and ciliates



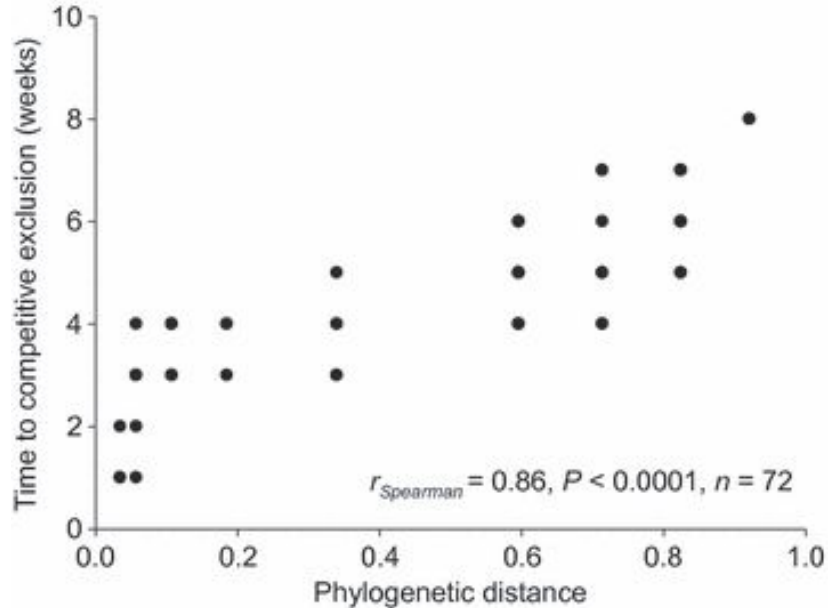
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Trait analysis suggests mouth size might play a role

Competition and ciliates



LETTER

Phylogenetic limiting similarity and competitive exclusion

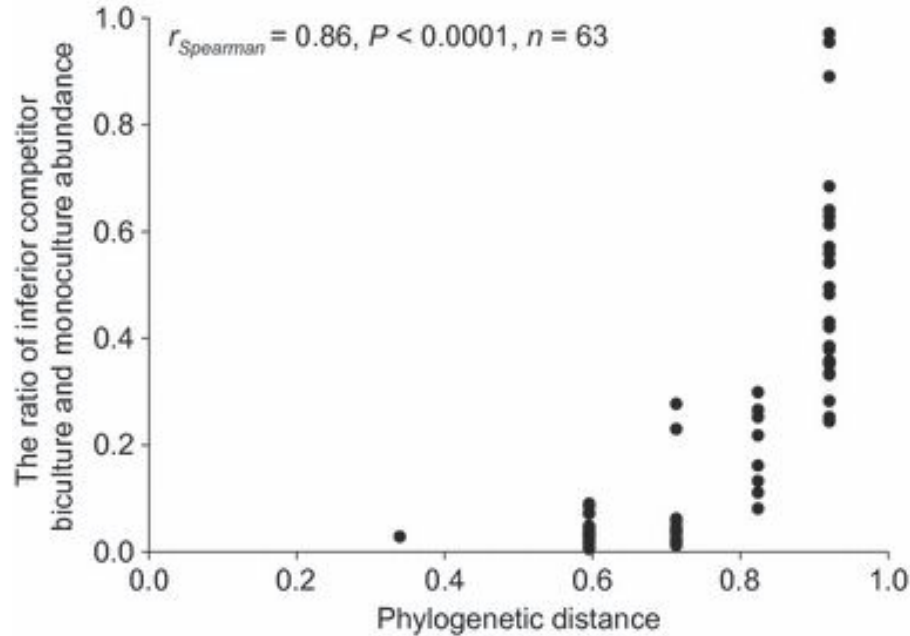
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Competitive exclusion happens faster with close relatives

Competition and ciliates



LETTER

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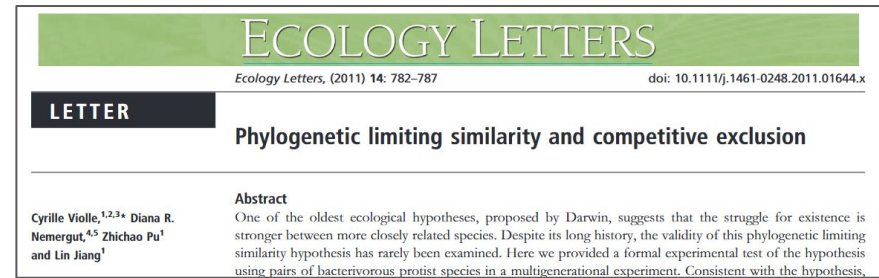
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Smaller effects relative to monoculture at larger distances

Competition and ciliates



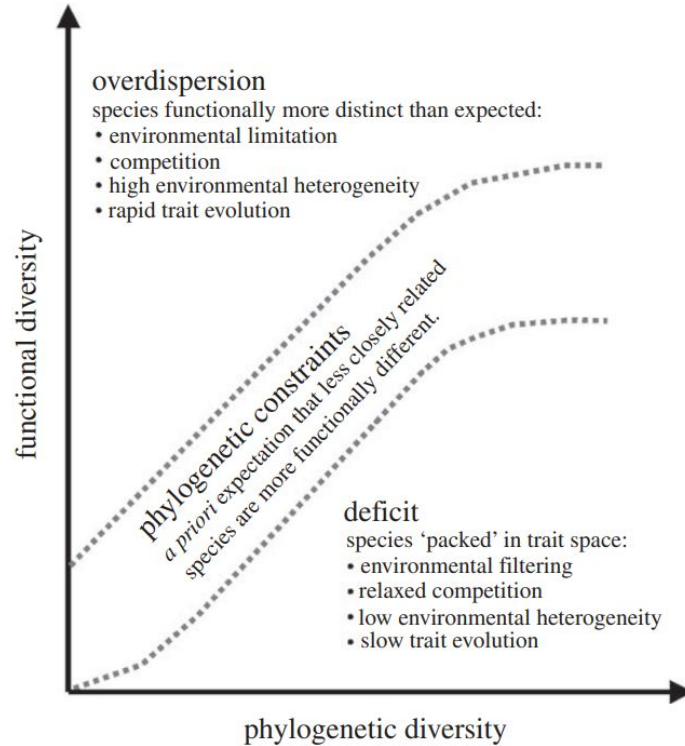
Provides strong, experimental test of competition driving assembly of ciliate communities

- Phylogeny served as a useful predictor
- Seems to be driven by feeding niches

Close relatives

- Faster exclusion
- More likely to exclude each other
- Larger impact on population sizes

Mammal assembly



Research

Understanding global patterns of mammalian functional and phylogenetic diversity

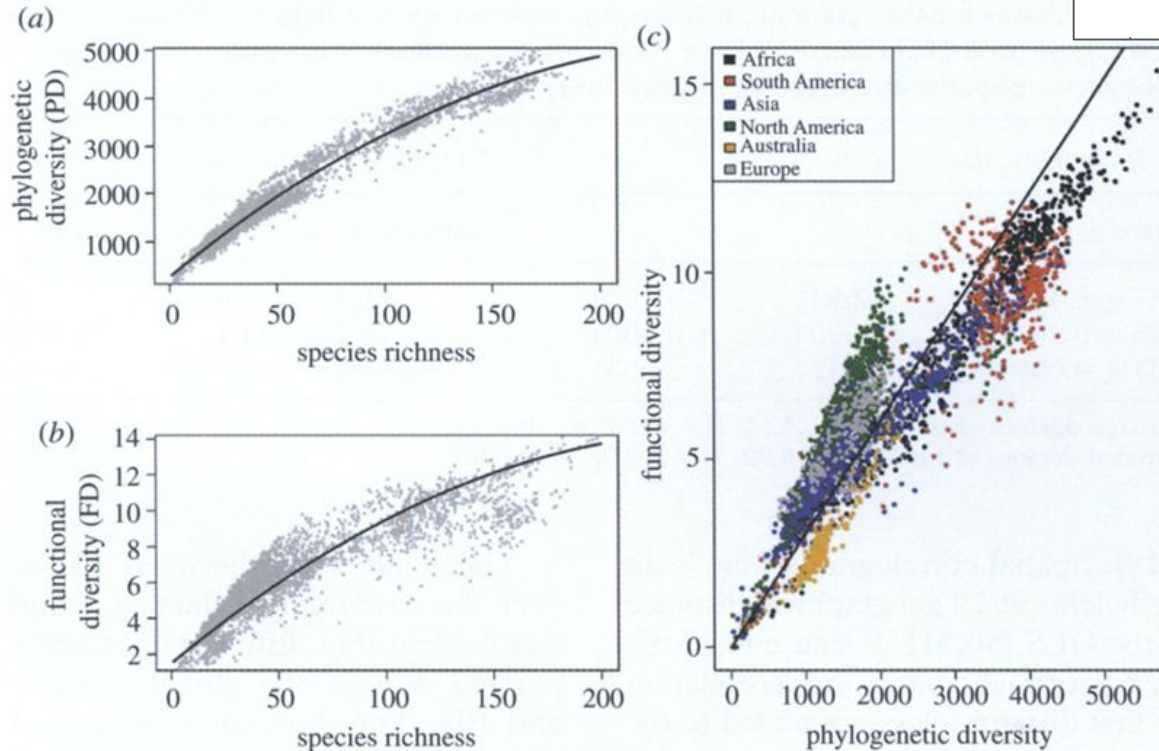
Kamran Safi¹, Marcus V. Cianciaruso^{2,*}, Rafael D. Loyola²,
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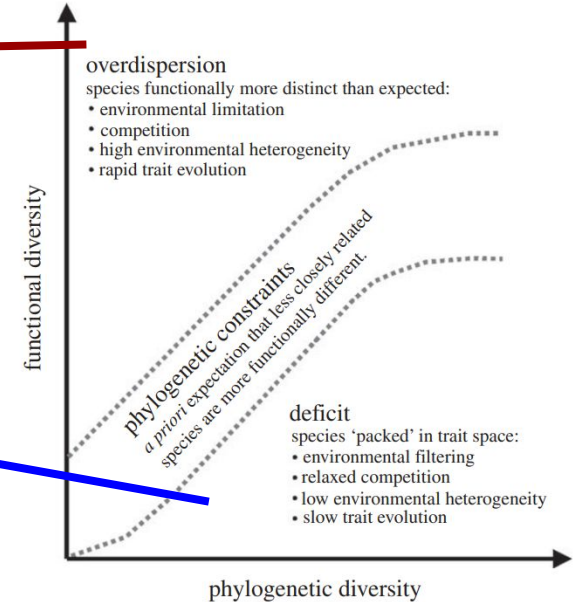
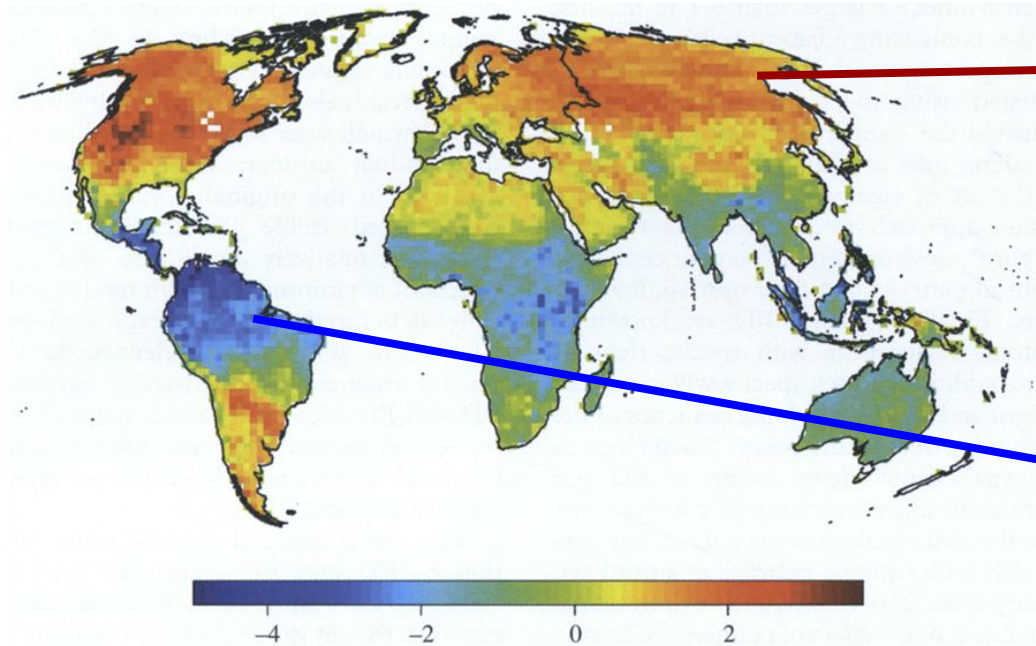


Mammal assembly

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Disease dynamics

LETTER

Past is prologue: host community assembly and the risk of infectious disease over time

Fletcher W. Halliday,^{1,2*}
Robert W. Heckman,¹
Wilfahrt³ and Charles E.
Mitchell^{1,3}



Abstract

Infectious disease risk is often influenced by host diversity, but the causes are unresolved. Changes in diversity are associated with changes in community structure, particularly during community assembly; therefore, by incorporating change over time, host community assembly may provide a framework to resolve causation. In turn, community assembly can be driven by many processes, including resource enrichment. To test the hypothesis that community assembly causally links host

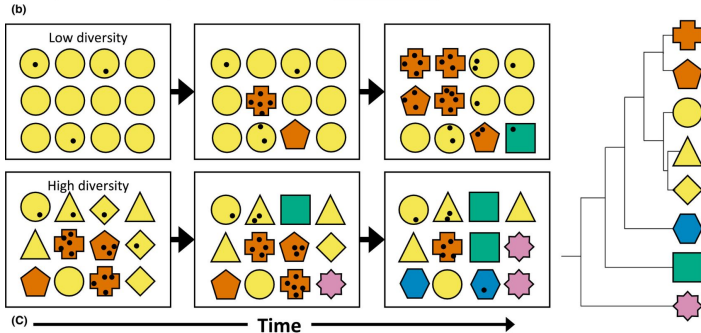
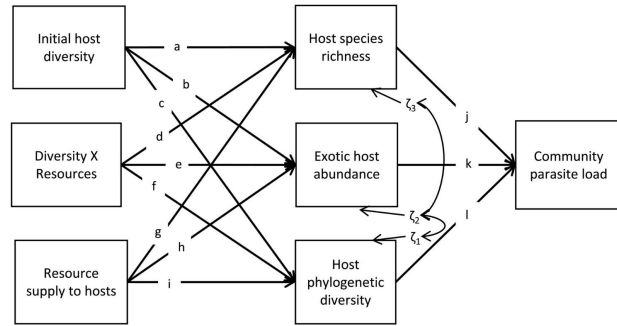
Disease dynamics



Biotic and abiotic drivers
(Established 2012)

Post-assembly host community structure
(Measured 2012, 2013, 2014)

Post-assembly disease
(Measured 2012, 2013, 2014)



LETTER

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Disease dynamics

Methods:

Hosts = plants

120 plots (1m x 1 m)

Treatments:

1 vs 5 species (6 different treatment sets of species each)

High vs low resource supply (fertilized vs unfertilized)

Disease:

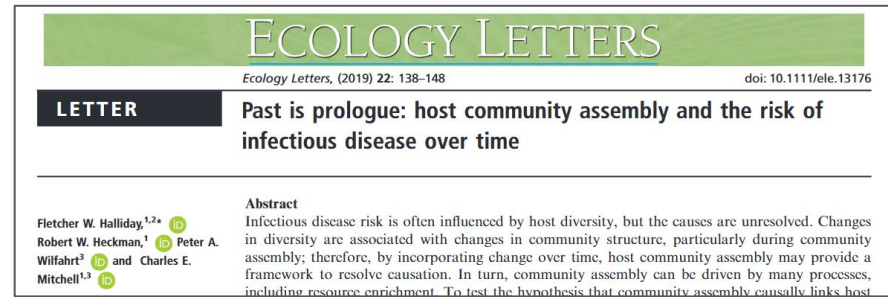
Subsets of leaves sampled per plot per species

Parasites identified by symptoms, DNA

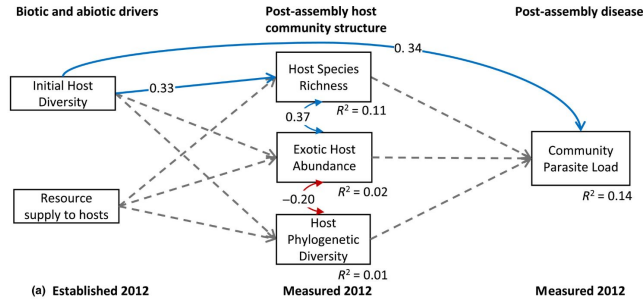
Calculated parasite load as total leaf area damaged per plot

Native vs Exotic

Followed USDA categorization



Disease dynamics



LETTER

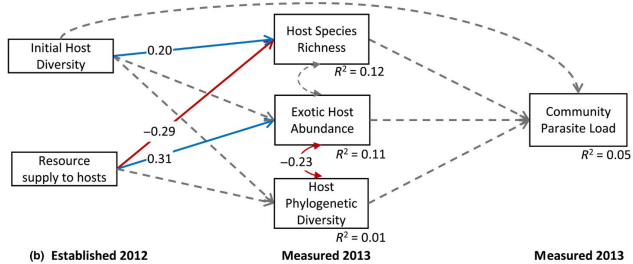
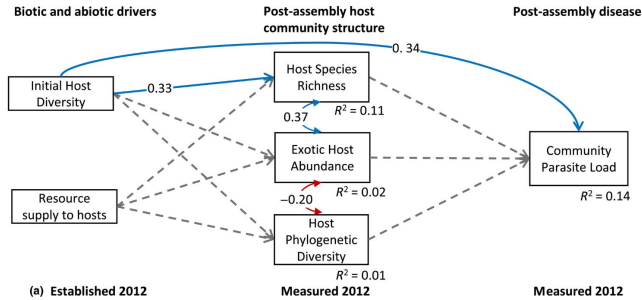
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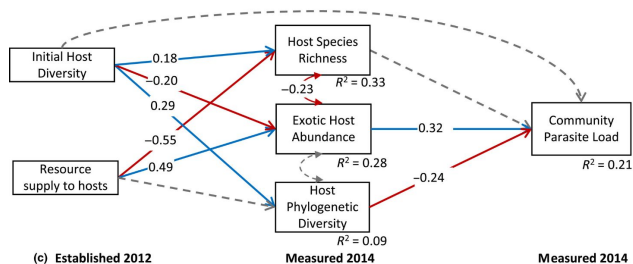
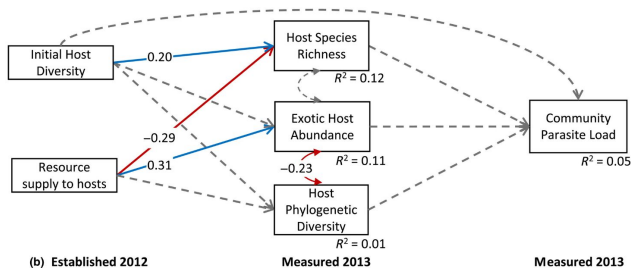
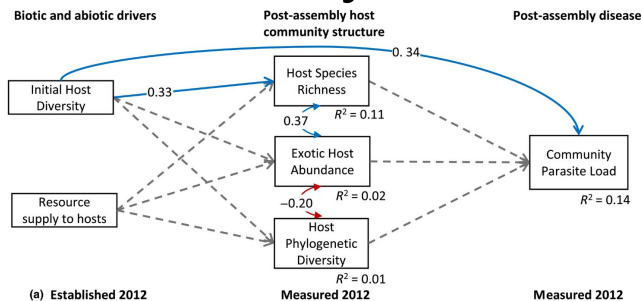
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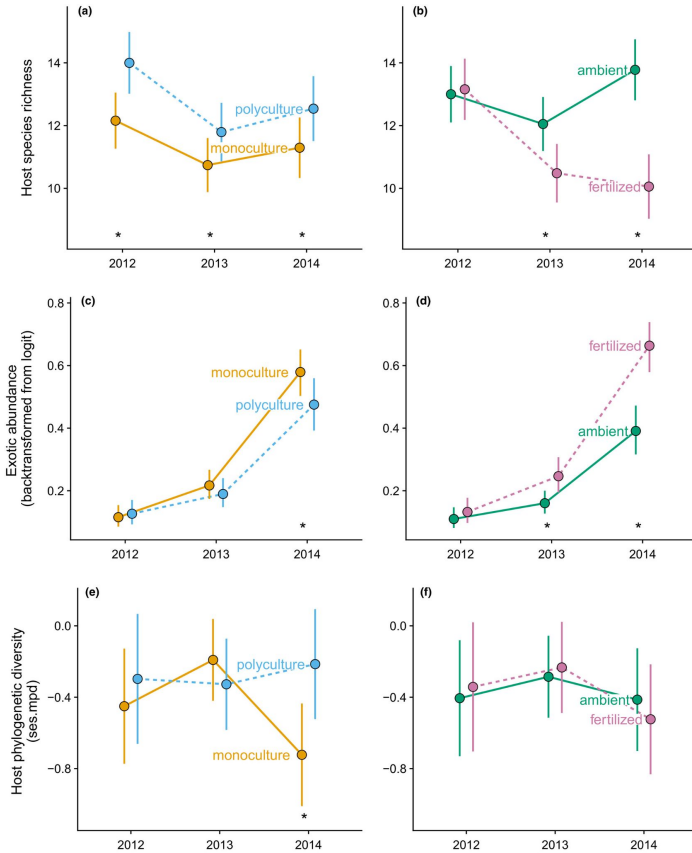
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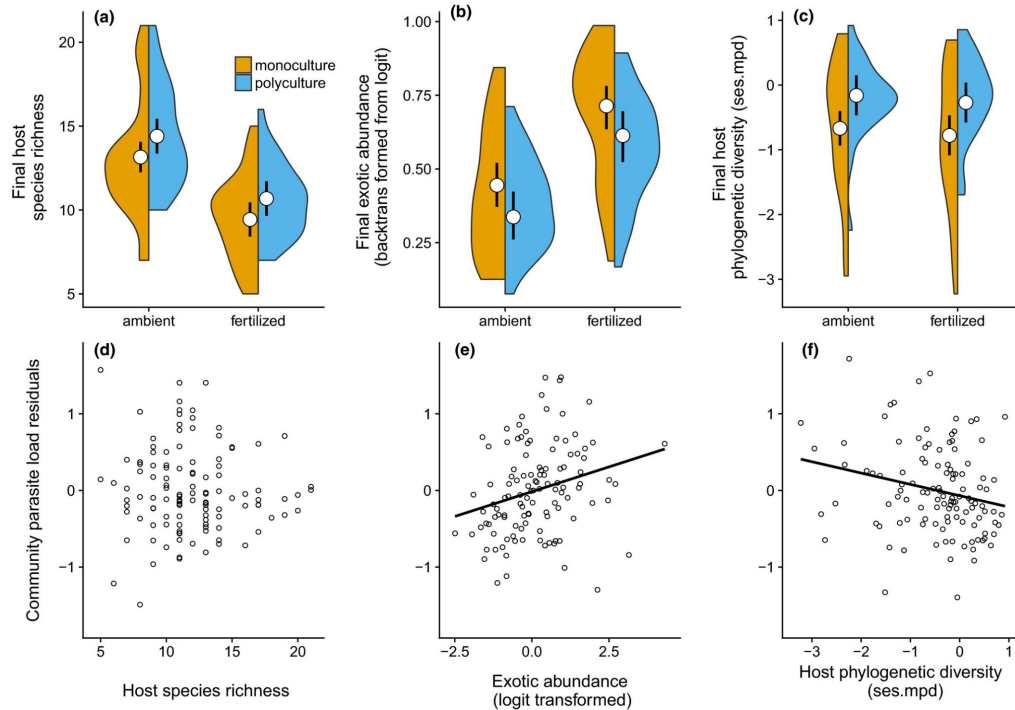
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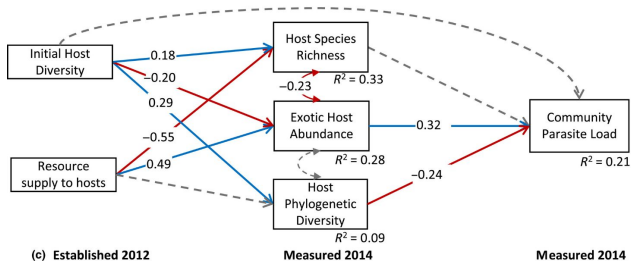
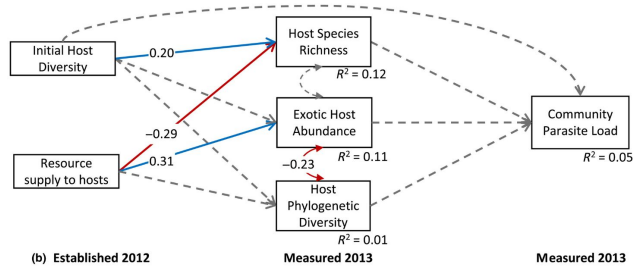
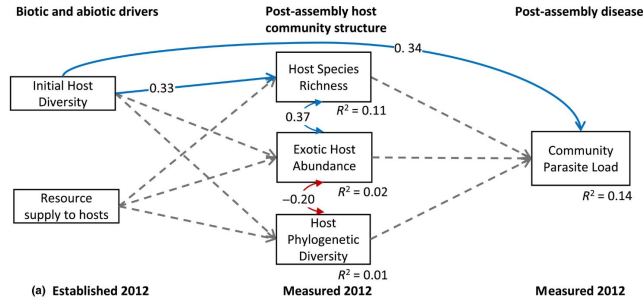
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Community assembly mechanisms can change over time!

- Importance of considering temporal dynamics.
- Just because you don't find evidence of something doesn't mean it won't be important at another time!

Next class: Community assembly and global change